

Running an End-to-End Machine-Learning Data Assimilation System on a Laptop: Porting HealDA to Apple Silicon, Auditing Its Numerical Trustworthiness, and Ablating GNSS Radio Occultation Observations

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Abstract

Machine-learning (ML) weather models now rival operational numerical weather prediction (NWP), and end-to-end ML data assimilation (DA) systems such as HealDA (Gupta et al. 2026) can map a 24-hour window of observations directly to a global analysis state in a single forward pass. Yet such systems are normally evaluated only on data-centre GPUs, and little is known about (i) whether they can run, reproducibly and trustworthily, on consumer accelerator hardware, and (ii) how the composition of the observation stream shapes the resulting analysis. We port HealDA (327 M parameters, HEALPix ViT) to an Apple M5 laptop with 32 GB of unified memory using the PyTorch Metal Performance Shaders (MPS) backend, requiring two non-trivial adaptations: a vendored `earth2grid` shim for HEALPix indexing, and a patch suite for two Metal-specific defects—a hard crash in zero-input `nn.Linear` layers of the `adaLN-Zero` conditioning path, and, more insidiously, silent corruption of large `int64` host-to-device copies during observation encoding. We further show that the corruption is *not* a precision problem (the model runs in `bfloat16` autocast on both CPU and MPS, and `fp32` MPS agrees with `fp32` CPU to 7×10^{-7}), but a memory-safety defect that only manifests at full observation scale (1.1×10^7 observations) and mimics a scientific error: without the fix, the zonal wind

of the analysis is essentially decorrelated from ERA5 ($r = -0.01$), while with the fix CPU and MPS analyses agree to `bf16` noise (z_{500} RMSE $3.9 \text{ m}^2 \text{ s}^{-2}$, $r = 1.0000$). Using the repaired pipeline we evaluate analyses against ERA5 over East Asia for 2024-06-01 00 UTC: adding AMSU-A/ATMS/MHS radiances (3.65×10^6 values) to the conventional stream (1.13×10^7 values) reduces layer-mean RMSE by 68% (temperature), 78% (geopotential), 36% (humidity), 36% (zonal wind) and 20% (meridional wind), and removes a -25 to -30 gpm zonal-mean bias in z_{500} . A four-way ablation of GNSS radio occultation (RO) observations—which constitute 71% of the conventional stream by count—shows that (a) RO contributes 10–15% additional RMSE reduction in temperature, geopotential and humidity over an RO-free conventional analysis, (b) RO alone matches or exceeds the full conventional stream in the upper stratosphere (150 hPa temperature $r = 0.979$ vs. 0.974) but degrades sharply below 500 hPa, and (c) removing the two background-value channels (`gps_t`, `gps_q`) that HealDA ingests alongside bending angles collapses the meridional-wind skill of an RO-only analysis from $r = 0.44$ to $r = 0.05$, demonstrating that the network has learned a joint $O-B$ mapping rather than a pure retrieval. We discuss the implications for ingesting non-UFS RO data such as FengYun-3 GNOS into a pretrained HealDA, and for the general practice of verifying accelerator-ported ML DA systems.

*Code and data: <https://github.com/sonet-hkust/healda-apple-silicon>. The manuscript was prepared with the assistance of an AI coding agent (WorkBuddy); all experiments were run on the author’s local machine.

1 Introduction

The skill of AI weather models has advanced explosively, from FourCastNet (Pathak et al. 2022) and Pangu-Weather (Bi et al. 2023) to GraphCast (Lam et al. 2023) and Aurora (Bodnar et al. 2024), with medium-range skill now comparable to operational NWP (Bauer et al. 2015). Almost all of these systems, however, still depend on a conventional variational DA system to provide initial conditions, which ties an otherwise near-instantaneous forecast pipeline to hours of high-performance computing. HealDA (Gupta et al. 2026) addresses this with an encoder–decoder transformer that maps a short window of satellite and conventional observations directly onto a global 1° analysis state on the HEALPix sphere (Górski et al. 2005), trained against ERA5 (Hersbach et al. 2020) using replayed UFS GEFSv13 observations. HealDA analyses initialized off-the-shelf forecast models (FCN3, Aurora, FengWu) lose less than one day of effective lead time relative to ERA5-initialized forecasts.

Two aspects of HealDA’s ecosystem remain underexplored. First, its public release has been exercised almost exclusively on CUDA GPUs; the vendor-locked `earth2grid` extension shipped as CUDA-12 x86_64 binaries, and the interaction of its observation encoder with non-CUDA backends is undocumented. Second, its sensitivity to the *composition* of the observation stream—which sensors matter, and how the network represents observation types it was trained on—is only partially characterized by the end-to-end scores in Gupta et al. (2026). Both questions matter for groups wishing to re-use HealDA with their own observation sources, for example the FengYun-3 GNOS radio occultation sounder (Liao et al. 2016), whose data are processed operationally in China but are not part of the UFS replay stream.

This paper reports a complete, reproducible local deployment of HealDA on Apple Silicon and uses it to run a series of controlled observation-subset experiments. Our contributions are:

1. **Port.** A minimal, non-invasive adaptation of HealDA to the PyTorch MPS backend on an Apple M5 (32 GB unified memory), including a pure-PyTorch shim for the CUDA-only `earth2grid` HEALPix grid, and two

`mips_patch` modules for Metal-specific defects (§3). The full conventional-observation analysis runs in 59 s (vs. 960 s on CPU) on a laptop.

2. **Trust audit.** We show that a large CPU–MPS analysis discrepancy is *not* a floating-point precision issue but a silent memory-corruption defect in Metal’s handling of large `non_blocking int64` copies, we localize it with layer-wise hook bisection, and we repair it with an input-construction patch whose correctness we verify down to bf16 noise (§4). We document the diagnostic path because the failure mode—plausible-looking but degraded analyses at scale, with identical results at toy scale—is a hazard for any back-end port of an ML DA system.
3. **Observation-subset evaluation.** Against ERA5 over East Asia, we quantify the analysis impact of (i) satellite radiances (AMSU-A, ATMS, MHS) on top of the conventional stream and (ii) GNSS radio occultation observations within it, via a four-way ablation (all conventional / conventional without RO / RO three-channel / RO bending-angle only) (§5, §6); the RO ablation is undertaken out of the author’s professional interest in GNSS-RO data assimilation.
4. **Interpretation.** We clarify the semantics of the RO channels in the UFS diagnostic stream—the companion `gps_t` and `gps_q` channels are *model-background samples at the bending-angle locations*, not retrievals—and connect the ablation outcome to the classical 3D-Var formalism (Lorenc 1986, Courtier et al. 1994, Daley 1991) and to the known prior-contamination issue of 1D-Var retrieval assimilation (§7).

All code, figure-generation scripts, per-level verification statistics and the compiled manuscript are publicly available¹.

¹<https://github.com/sonet-hkust/healda-apple-silicon>

2 The HealDA system and data

2.1 Architecture

HealDA (Gupta et al. 2026) is an observation encoder followed by a HEALPix vision transformer (a DiT-L adaptation with 24 blocks, width 1024, 327.2 M parameters). The encoder buckets observations by sensor family into fixed-size token grids; each (pixel, bucket) cell is aggregated with a masked scatter-mean so that absent sensors are explicitly flagged through an observability mask. The trunk operates on 12288 tokens (HPX64 grid, $n_{\text{side}} = 64$, 49152 cells at $\approx 0.92^\circ$, 2×2 patchification), and the decoder emits 74 output channels: five dynamical fields (temperature T , zonal and meridional wind u, v , geopotential z , specific humidity q) on 13 pressure levels (1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 hPa) plus nine surface variables. The release checkpoint (`healda_ufs_era5.mdlus`, 1.22 GB) carries five observation embedders: `conv` (8 conventional channels $\times$ 33-wide tokens = 264 bins) and four radiance buckets (AMSU-A 120 platform-channel bins, ATMS 44, MHS 25, AMSU-B). Conditioning is unconditional in the released configuration; the adaLN-Zero modulation layers are therefore `nn.Linear(0, 6144)`—a detail that becomes important in §3. Inference is a single forward pass over the 24-hour observation window $[t_0 - 21\text{h}, t_0 + 3\text{h}]$.

2.2 Observation stream

We use the NOAA UFS GEFSv13 replay observations served by `earth2studio` (NVIDIA 2025) for 2024-06-01 00 UTC. Table 1 inventories the stream. Two facts deserve emphasis. First, RO bending angles and their two companion channels constitute 8.04×10^6 of the 1.134×10^7 conventional values (71%); HealDA’s conventional bucket is dominated, by count, by GNSS RO. Second, the RO companion channels `gps_t` and `gps_q` are not retrievals: in the UFS/GSI diagnostic convention they are the *model first-guess temperature and specific humidity sampled at the bending-angle report locations*, while the BUFR 1D-Var retrieval fields (descriptors 12001/13001) are deliberately not exported. We return to the

Table 1: Observation inventory for 2024-06-01 00 UTC (window $[t_0 - 21\text{h}, t_0 + 3\text{h}]$, UFS GEFSv13 replay). “RO” = GNSS radio occultation.

Group	Channel	Count
RO (bending angle)	<code>gps</code> (α , rad)	2,680,664
	<code>gps_t</code> (background T , K)	2,680,664
	<code>gps_q</code> (background q)	2,680,664
In-situ / profiles	u	1,080,489
	v	1,080,489
	T	617,321
	surface pressure	428,354
	q	91,356
Conventional subtotal		11,340,001
Radiances	AMSU-A	2,136,601
	ATMS	985,798
	MHS	528,756
Satellite subtotal		3,651,155

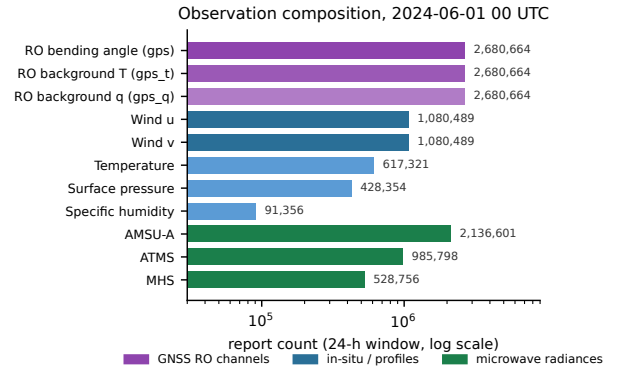


Figure 1: Observation composition of the 2024-06-01 00 UTC stream (counts of Table 1, logarithmic axis). The three GNSS-RO channels dominate the conventional bucket, and AMSU-A dominates the radiance suite.

implications in §7.

The reported bending angles have median 8.96×10^{-4} rad, median tangent pressure 30.4 hPa, and geographic coverage to $\pm 79^\circ$ latitude, consistent with the limb-sounding geometry of GNSS RO (Kursinski et al. 1997, Anthes et al. 2008). Figure 1 visualizes the inventory on a logarithmic count axis; the dominance of the three RO channels is immediate.

2.3 Verification data

Analyses are verified against the ERA5 pressure-level reanalysis (Hersbach et al. 2020) valid at the same time, interpolated from a 0.25° regional extract ($93\text{--}127^\circ\text{E}$, $16\text{--}43^\circ\text{N}$; East Asia) onto the common 13-level set. HealDA fields are sampled from the native HEALPix grid onto the ERA5 grid by Delaunay linear interpolation with nearest-neighbour infill. We report RMSE, the Pearson correlation r with ERA5, and the ratio of spatial standard deviations (a sharpness indicator). A single case study cannot establish climatological significance; its value is process-level attribution under fully controlled input subsets, enabled by the fact that a HealDA analysis is a deterministic function of its observation vector.

3 Porting HealDA to Apple Silicon

3.1 Environment

The host is an Apple M5 laptop with 32 GB unified memory running macOS. The software stack is Python 3.12.12 (managed by uv), PyTorch 2.10.0 with the MPS backend, and the NVIDIA earth2studio 0.18.0 / physicsnemo 2.2.2 releases with timm 1.0.30 and makani 0.2.0. No CUDA components are installed.

3.2 The earth2grid shim

HealDA’s input pipeline indexes observation locations onto the HEALPix grid through NVIDIA’s earth2grid package, whose official distribution ships CUDA-12 x86_64 binaries and therefore cannot be installed on Apple Silicon. The HEALPix functionality actually required by the released checkpoint is narrow: an `ang2pix` mapping from (longitude, latitude) to pixel index under the padded XY pixel ordering, plus grid lat/lon attributes. We vendor a ~ 100 -line pure-PyTorch shim implementing this subset, validated to reproduce the reference pixel indexing. The shim is built on demand by a small `build_shim.py` utility and inserted at the front of `sys.path`; the upstream package is never modified.

3.3 Metal defect 1: zero-input `nn.Linear`

Because the released HealDA is an unconditional regressor, the adaLN-Zero modulation layers are degenerate `nn.Linear` instances with `in_features = 0` (25 of them: 24 blocks plus the detokenizer). On MPS, a forward pass through such a layer aborts the process inside `MPSNDArray initWithDevice:...` with `Error: device may not be nil` (SIGABRT without a Python traceback); merely moving the layer to Metal is harmless, only the forward crashes.

The repair exploits the fact that with zero inputs the layer is a *constant function*:

$$y = xW^\top + b = b, \quad (1)$$

so evaluating it anywhere is exact. `patch_for_mps()` replaces each such layer with a CPU-pinned stub whose `_apply` is a no-op (so a later `.to("mps")` cannot drag the weights onto Metal) and whose forward simply returns the stored bias. Output equality with the original layer is bit-exact ($\max |\Delta| = 0.00$ over the full analysis run).

3.4 Metal defect 2: silent corruption of large `int64` copies

The second defect produces no crash, no warning and no Python-level error—it silently degrades the science. It is the subject of the next section.

4 A numerical trust audit of the MPS backend

4.1 Symptom

The first MPS analyses (2024-06-01 00 UTC, full conventional stream) verified far worse against ERA5 than a CPU reference run of the same configuration: layer-mean r of meridional wind was -0.01 (i.e. uncorrelated), zonal wind 0.45, and temperature 0.75, versus 0.88, 0.94 and 0.95 on CPU (Table 2). Initial suspicion fell on reduced-precision arithmetic.

4.2 The precision hypothesis is false

Precision was ruled out by direct measurement:

- The inference script hard-codes bfloat16 autocast on *both* devices (`torch.autocast(self.device.type, dtype=torch.bfloat16)`); the CPU reference was never an fp32 golden run. With the autocast op-routing conventions verified identical on CPU and MPS (matmuls and linear layers in bf16; softmax, layer-norm and reductions in fp32), a 24-layer LayerNorm-plus-residual chain at width 1024 diverges from an fp32 reference by 4.81×10^{-3} on CPU and 4.81×10^{-3} on MPS—indistinguishable—while MPS fp32 agrees with CPU fp32 to 7.2×10^{-7} .
- The masked scatter-mean aggregation (`scatter_reduce_` with `reduce="mean"`) is *bit-identical* between CPU and MPS at every occupancy tested (2–1000 observations per cell, 0.00e+00 difference), in both fp32 and bf16.
- The 25 CPU-pinned stubs of §3 return exactly the original biases ($\max |\Delta| = 0.00$).

Notably, a 20 000-observation subsample produced *bit-consistent* encoder outputs between devices (relative $\ell_2 = 8.6 \times 10^{-7}$): the defect scales with observation count, the signature of a resource-management bug rather than arithmetic.

4.3 Localization

Registering forward hooks on every stage of the observation encoder and bisecting the CPU–MPS divergence localized the corruption to the input-construction routine `build_input()`, *before* any network computation: of its outputs, `obs`, `pix`, `obs_type` and `offsets` matched CPU exactly, while the metadata tensor was already garbage on MPS. A targeted re-check identified the corrupted tensor:

```
target_time (shape [1.01 × 107], int64):
CPU: 1717200000, 1717200000, ... (epoch
seconds, correct)
MPS: 17171244000000000000, ... (observed
times in nanoseconds)
```

`build_input()` assembles its tensors column-by-column with `.to("mps",`

`non_blocking=True)`. Under Metal, the later copies in this sequence overwrite earlier buffers—a buffer-lifetime / aliasing defect in the host-to-device copy path—so the `target_time` column arrives corrupted with values belonging to a different column. Downstream, the relative-time feature computes $\Delta t = (\text{target_time} \times 10^9 - t_{\text{obs}})$, which overflows `int64` and yields 10^8 -second garbage; the encoder is then fed wrong timing information for every observation. Calling `torch.mps.synchronize()` does not repair it, and a minimal isolated reproducer with a single `non_blocking` copy does not trigger it, indicating sensitivity to the specific multi-column copy sequence rather than to any single call.

4.4 Repair and verification

The fix, `patch_build_input_on_cpu()`, temporarily swaps `model.device_buffer` for an empty CPU tensor so that `model.device` reports CPU and `build_input()` constructs its tensors entirely in host memory; the completed dictionary is then moved to the accelerator with synchronous copies. Observation encoding becomes device-consistent (relative $\ell_2 = 1.99 \times 10^{-2}$, $r = 0.99979$ vs. CPU; the residual is bf16 noise), and the analysis is restored (Table 2, Figs. 2–3). Two further checks support stability: two independent MPS runs differ by less than CPU and MPS differ (0.015 K vs. 0.033 K layer-mean temperature RMSE), i.e. no stochastic corruption remains; and the conventional+satellite run reproduces the same behaviour (406 s, 10.5 GB peak).

5 Conventional versus conventional-plus-satellite

With the repaired pipeline we quantify the analysis impact of microwave radiances by comparing the conventional-only analysis with one that additionally ingests 3.65×10^6 AMSU-A/ATMS/MHS radiance values (Table 2, Figs. 5–8).

Radiance impact is large across the column. Layer-mean RMSE falls by 68% for temperature (2.50 to 0.80 K), 78% for geopotential (612 to $135 \text{ m}^2 \text{ s}^{-2}$), 36% for specific humidity, 36% for zonal wind and 20% for meridional wind;

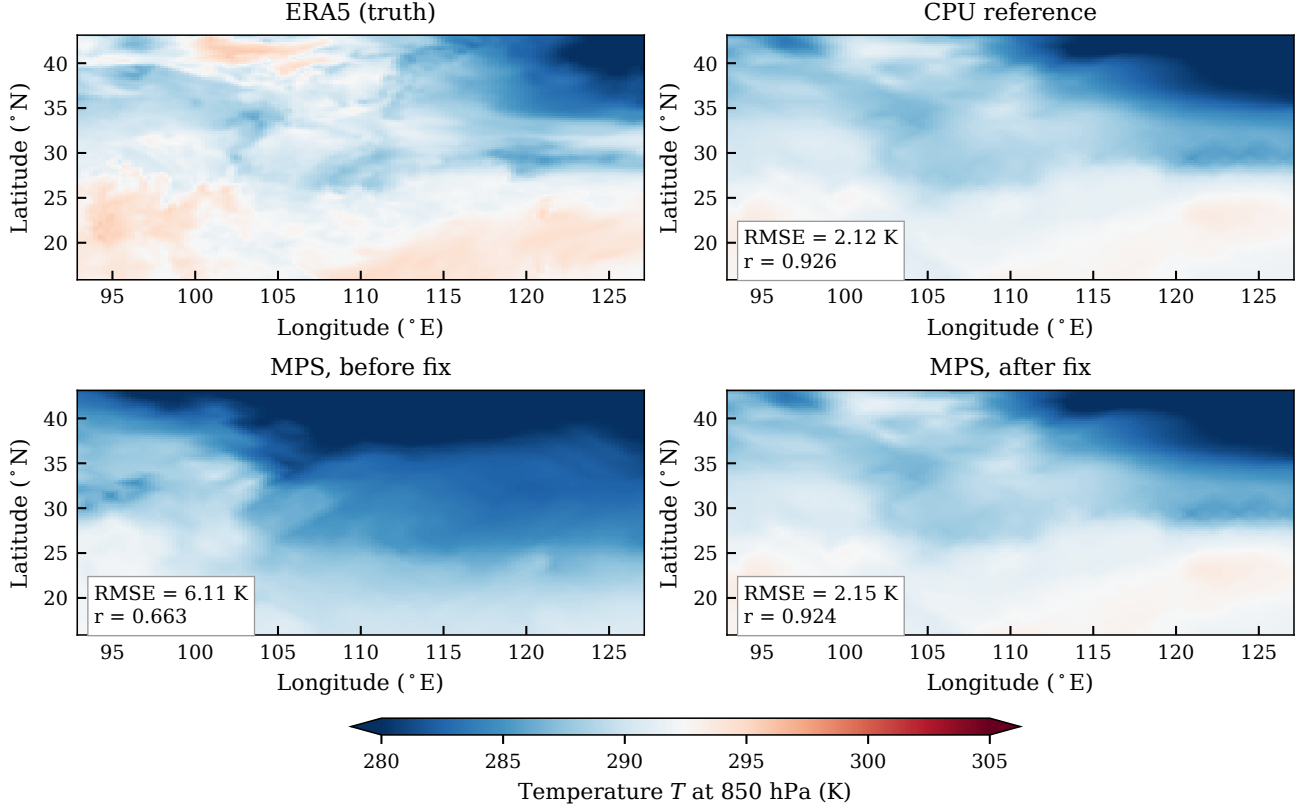


Figure 2: 850 hPa temperature over East Asia for 2024-06-01 00 UTC. Top: ERA5 truth and the CPU reference analysis. Bottom: the MPS analysis before and after the input-construction repair. Before the repair the MPS analysis is smoothly biased (RMSE 6.11 K, $r = 0.663$); after the repair it matches the CPU reference (RMSE 2.15 K vs. 2.12 K, $r = 0.924$ vs. 0.926).

Table 2: Layer-mean verification against ERA5 over East Asia (13 levels, 2024-06-01 00 UTC). RMSE in K (T), m s^{-1} (u, v), $\text{m}^2 \text{s}^{-2}$ (z) and mg kg^{-1} (q); r = spatial correlation; SR = standard-deviation ratio.

Experiment	T		u		v		z		q	
	RMSE	r	RMSE	r	RMSE	r	RMSE	r	RMSE	r
CPU reference (bf16)	2.50	0.946	3.03	0.943	2.42	0.880	611	0.960	592	0.844
MPS before repair	5.00	0.749	8.76	0.454	7.48	-0.011	1431	0.746	2155	0.323
MPS after repair (conv)	2.50	0.946	3.04	0.943	2.41	0.881	612	0.960	590	0.844
Conventional + satellite	0.80	0.983	1.96	0.967	1.92	0.907	135	0.993	378	0.923

Table 3: Cost of a single 2024-06-01 00 UTC analysis on the M5 laptop (32 GB unified memory).

Configuration	Device	Wall time	Peak memory
Conventional	CPU	960 s	16.0 GB
Conventional	MPS	59 s	8.6 GB
Conv. + satellite	MPS	406 s	10.5 GB

correlations rise to 0.98–0.99 for T and z everywhere below 100 hPa. At 850 hPa, temperature

RMSE drops from 2.15 to 0.74 K and r from 0.924 to 0.981.

The largest gains are tropical. In the horizontal (Fig. 6), the conventional analysis shows its worst temperature and geopotential errors in the 16–25°N monsoon region (up to 6 K at 850 hPa, $-600 \text{ m}^2 \text{s}^{-2}$ at 500 hPa), where the conventional network is sparse; these are nearly eliminated by the radiances. Mid-latitude wind error changes little, and isolated small patches degrade slightly—consistent with the radiance-suite

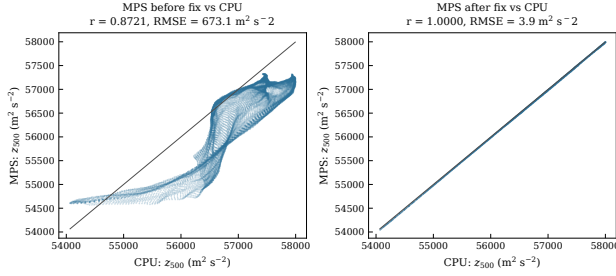


Figure 3: 500 hPa geopotential, MPS vs. CPU reference, over all verification points. Left: before the repair ($r = 0.872$, $\text{RMSE} = 673 \text{ m}^2 \text{ s}^{-2}$). Right: after the repair ($r = 1.0000$, $\text{RMSE} = 3.9 \text{ m}^2 \text{ s}^{-2}$, consistent with bf16 noise).

weighting being optimized (at training time) for global rather than regional skill.

The increment itself is positive and broad-scale. Figure 4 shows the raw analysis increment, $\text{conv} + \text{sat} - \text{conv}$, at 500 hPa. Both temperature and geopotential increments are positive almost everywhere over East Asia, with z_{500} lifted by roughly $100\text{--}400 \text{ m}^2 \text{ s}^{-2}$ ($\approx 10\text{--}40 \text{ gpm}$) across the domain—the signature of the radiances pulling the analysis upward toward ERA5, consistent with the bias removal quantified next.

The mass-field bias is removed. The conventional analysis carries a zonal-mean z_{500} bias of -25 to -30 gpm across $15\text{--}43^\circ \text{N}$ (Fig. 8); with radiances the bias is within -8 to -5 gpm . We caution that a single-window ML DA system has no cycling background: part of this bias is attributable to the absence of a dynamical background rather than to the missing observations, and the radiance increment should be read as the envelope of both effects.

6 Ablation of GNSS radio occultation observations

6.1 Design

Within the conventional bucket we ran a four-way ablation, enabled by the per-(pixel, bucket) observability masks with which HealDA encodes “this sensor family is absent here”:

1. **Conv (all):** the full conventional stream (1.134×10^7 values, 71% of them RO-related);

2. **Conv w/o RO:** the same stream with the three RO channels removed (`--conv-vars pres q t u v`);
3. **RO 3-channel:** bending angles plus their background T/q companions only (`gps_t gps_q`);
4. **RO bending-angle only:** `gps` alone.

All four analyses use the repaired MPS pipeline and are verified identically.

6.2 Results

Table 4 summarizes layer-mean statistics; Figs. 10 and 12 show vertical and horizontal structure.

RO contributes 10–15% to the conventional analysis in mass, temperature and humidity, and little in wind. Removing RO degrades layer-mean RMSE by 10.2% (T), 12.3% (z) and 15.1% (q), but only 4.9–5.2% (u, v). This is the expected physics: bending angles constrain temperature and geopotential directly (through refractivity and hydrostatic balance) but influence winds only indirectly through geostrophic adjustment, mirroring the experience of variational RO assimilation (Cucurull et al. 2008, Healy and Thépaut 2006). The zonal-mean mass-field signature of the ablation is shown in Fig. 9: the -25 to -30 gpm bias of the full conventional analysis is barely changed by removing RO (noRO), but grows monotonically as the observation set is reduced to RO-only variants.

RO alone is competitive in the upper stratosphere and weak in the lower troposphere. At 150 hPa the RO-only (3-channel) temperature correlation, 0.979, exceeds the full conventional analysis (0.974); at 100 hPa it matches it (0.9689 vs. 0.9690) with a *lower* RMSE (3.34 vs. 4.37 K). Below 500 hPa the RO-only analysis degrades rapidly (850 hPa T : $r = 0.843$ vs. 0.924; Fig. 10), reproducing the classical “strong above, weak below” vertical signature of RO (Anthes et al. 2008, Healy and Thépaut 2006) within an ML DA system. Notably, the bending-angle-only analysis has the *lowest* 100 hPa temperature RMSE of all experiments (2.40 K) but the lowest standard-deviation ratio (0.73), i.e. a locally accurate but over-smoothed field—a reminder that

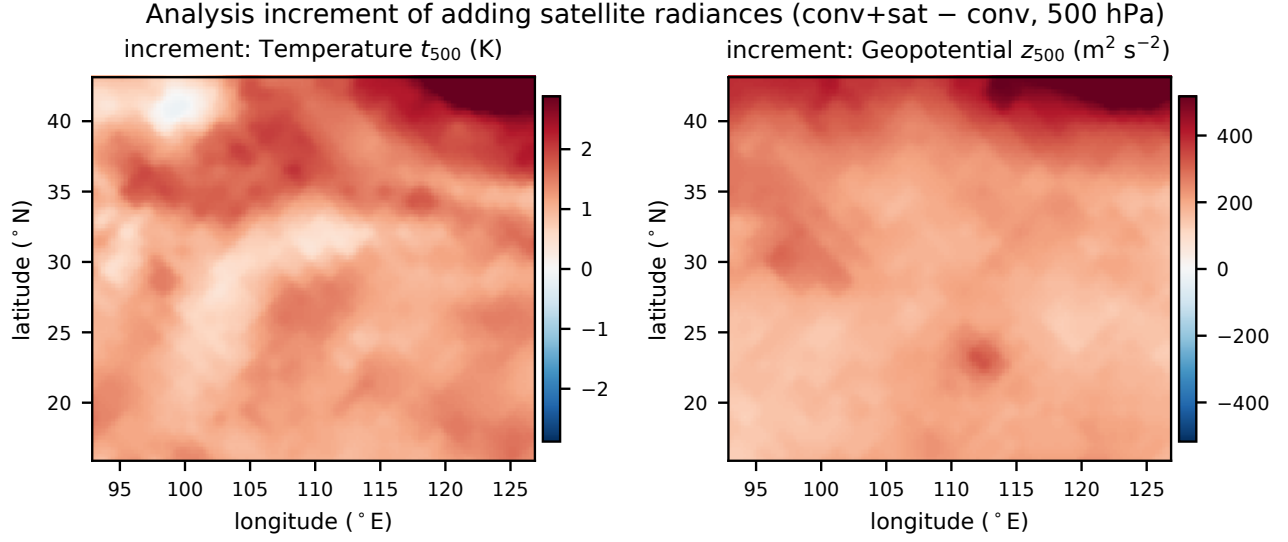


Figure 4: Direct analysis increment of adding satellite radiances (conv+sat – conv) at 500 hPa over East Asia: temperature (left, K) and geopotential (right, $\text{m}^2 \text{s}^{-2}$). The increment is positive nearly everywhere, lifting the mass field toward ERA5.

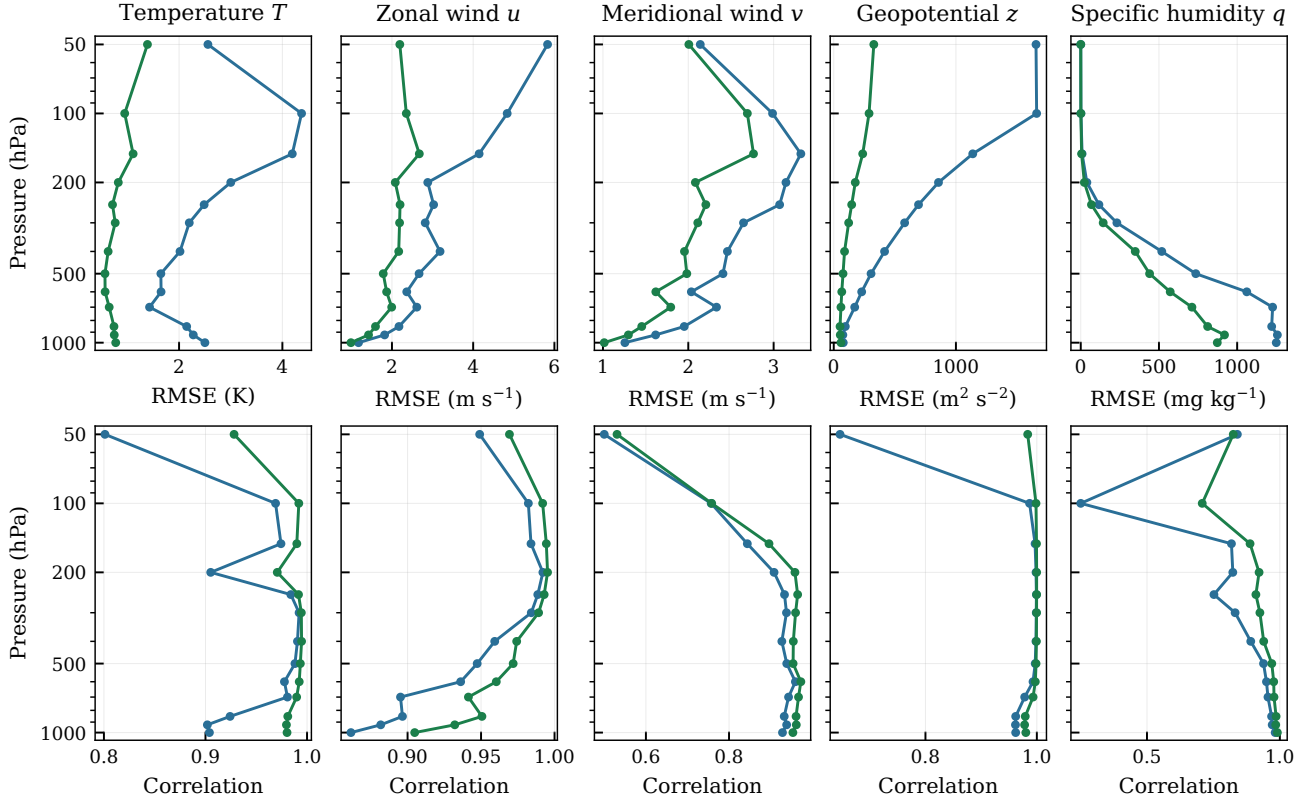


Figure 5: Vertical profiles of verification statistics against ERA5 (East Asia, 13 levels): RMSE (top) and spatial correlation (bottom) for the conventional-only and conventional-plus-satellite analyses.

RMSE alone can hide the loss of variance in a regression toward the mean. The same structure

is visible in the latitude–height cross-sections of Fig. 11: the cold bias that fills the troposphere

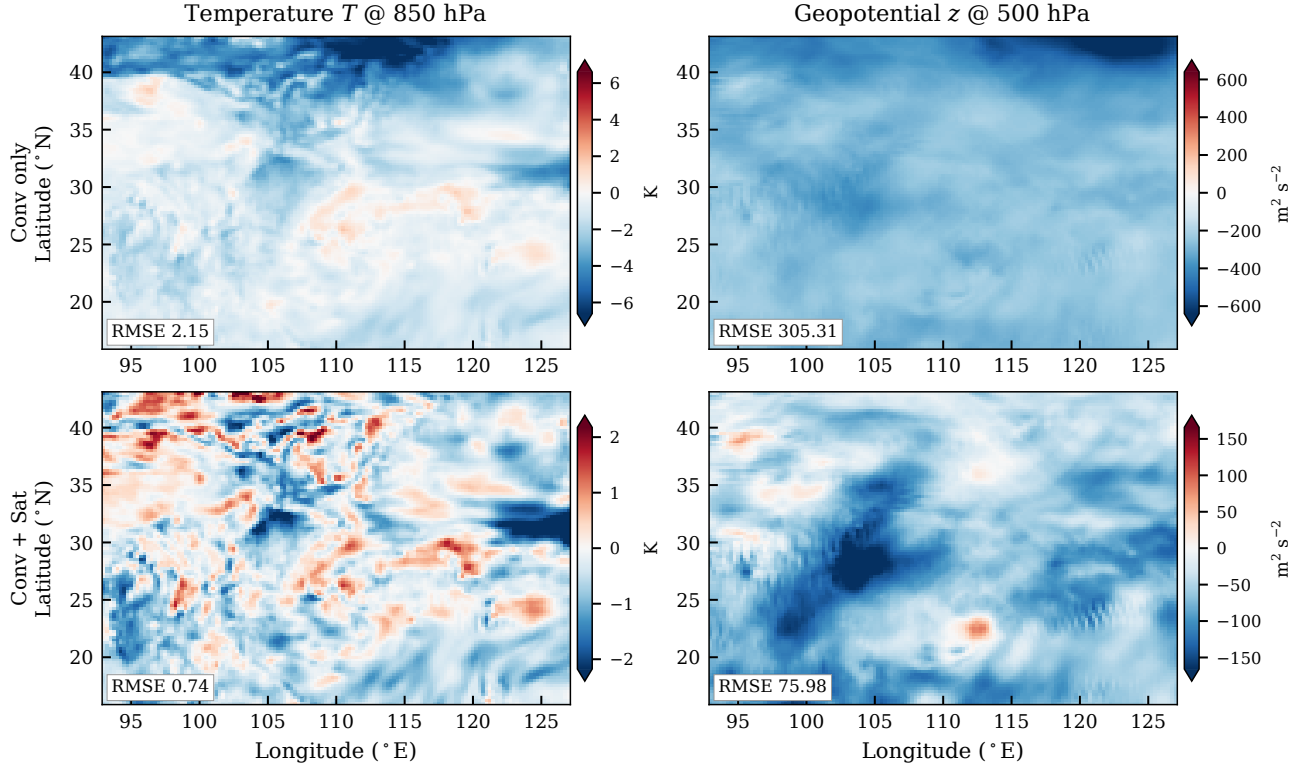


Figure 6: Analysis-minus-ERA5 differences for 850 hPa temperature (left) and 500 hPa geopotential (right), conventional-only (top) versus conventional-plus-satellite (bottom).

Table 4: GNSS-RO ablation: layer-mean verification against ERA5 (13 levels, East Asia). The “RO impact” column is the layer-mean RMSE reduction of *Conv (all)* relative to *Conv w/o RO*.

Experiment	T		u		v		z		q	
	RMSE	r	RMSE	r	RMSE	r	RMSE	r	RMSE	r
Conv (all)	2.50	0.946	3.04	0.943	2.41	0.881	612	0.960	590	0.844
Conv w/o RO	2.78	0.932	3.20	0.938	2.55	0.864	697	0.962	695	0.825
RO 3-channel	3.95	0.898	5.03	0.748	5.40	0.441	1144	0.858	1119	0.737
RO bangle only	5.56	0.707	9.90	0.333	6.37	0.051	1528	0.704	1950	0.575
RO impact (RMSE)	−10.2%		−4.9%		−5.2%		−12.3%		−15.1%	

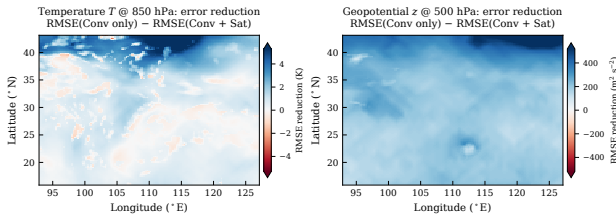


Figure 7: RMSE reduction (in absolute units) from adding satellite radiances, 850 hPa temperature (left) and 500 hPa geopotential (right). Blue: RMSE reduced; red: RMSE increased.

of the bending-angle-only analysis and the near-neutral full column of the three-channel analysis, in contrast to the structured, smaller-magnitude bias field of the full conventional stream.

Background companions carry decisive information. Dropping `gps_t/gps_q` from the RO-only analysis collapses layer-mean v from $r = 0.441$ to $r = 0.051$ (uncorrelated), T from 0.898 to 0.707, and z from 0.858 to 0.704. The network has not been trained to retrieve the atmosphere from bending angles; it has been trained to map the *pair* (observation, background) to an analysis.

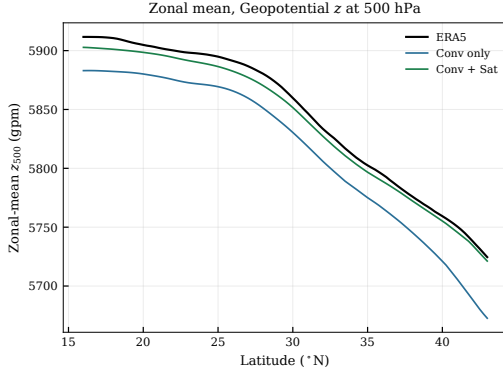


Figure 8: Zonal-mean 500 hPa geopotential height (gpm): ERA5 versus the conventional-only and conventional-plus-satellite analyses.

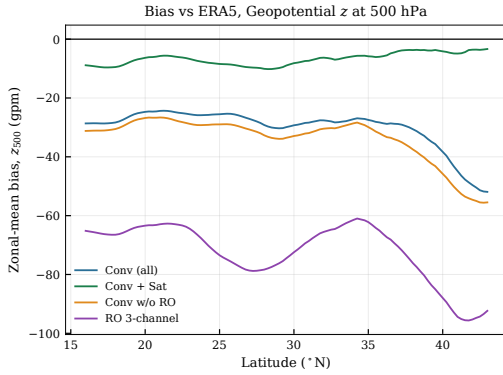


Figure 9: Zonal-mean z_{500} bias against ERA5 for the ablation experiments (plus the conventional-plus-satellite analysis for context).

We interpret this next.

7 Interpretation: RO channels as (O, B) pairs

In 3D-Var the RO analysis problem is

$$J = \frac{1}{2}(x - x_b)^\top B^{-1}(x - x_b) + \frac{1}{2}(H(x) - y_o)^\top R^{-1}(H(x) - y_o) \quad (2)$$

with increment $\delta x = BH^\top(HBH^\top + R)^{-1}d$ and innovation $d = y_o - H(x_b)$ (Lorenc 1986, Daley 1991, Courtier et al. 1994). The background enters *twice*: as the prior centre of the first term, and inside $H(x_b)$ in the innovation. The three RO channels of the UFS diagnostic stream correspond exactly to the pieces of this machinery that a learned operator must be fed explicitly:

Channel	3D-Var analogue	Status
<code>gps</code>	y_o (bending angle)	true observation
<code>gps_t</code>	$H(x_b)$, T component	background sample
<code>gps_q</code>	$H(x_b)$, q component	background sample

HealDA does not implement a bending-angle forward operator H ; it receives $H(x_b)$ directly (GSI diagnostics must store it to compute $O - B$ anyway) and learns the combined map $(y_o, x_b|_{\text{obs}}) \rightarrow x_a$, i.e. the roles that B , R , H^* (adjoint) and the innovation computation play in (2) are absorbed into the network weights. Three observations support this reading. First, the collapse experiment: without the B channels the RO-only analysis degenerates (§6), because the network has never seen bending angles unanchored by a background. Second, the geometry: a bending angle is a non-local Abel transform of the refractivity profile, $\alpha(a) = -2a \int_a^\infty \frac{d \ln n}{dx} \frac{dx}{\sqrt{x^2 - a^2}}$, whereas the network consumes un-ordered scatterer points that carry no profile identity—inverting the Abel integral from such a representation without a local anchor is severely ill-posed, which is exactly what the $r = 0.05$ meridional wind exhibits. Third, the observation-space semantics: the two background channels are kept separate from the (identical-unit) regular T and q channels—their normalization statistics differ by about 20 K in the mean, reflecting the stratosphere-weighted RO report distribution—and the released checkpoint’s own normalization file documents them as a distinct measurement method.

This design has an underappreciated advantage over the alternative operational pathway of assimilating 1D-Var retrievals: in retrieval assimilation the background used inside the retrieval leaks into the “observation” and is then counted again in (2) through the prior term, a double-counting that biases analyses toward the background and is a classical reason the community converged on bending-angle assimilation (Cucurull et al. 2008, Healy and Thépaut 2006). HealDA keeps the observation and the prior in separate, explicit channels—no prior is baked into the observation—so the information bookkeeping is cleaner. The cost is symmetrical: the influence of the background is frozen into the trained weights, with no external B -variance dial to turn,

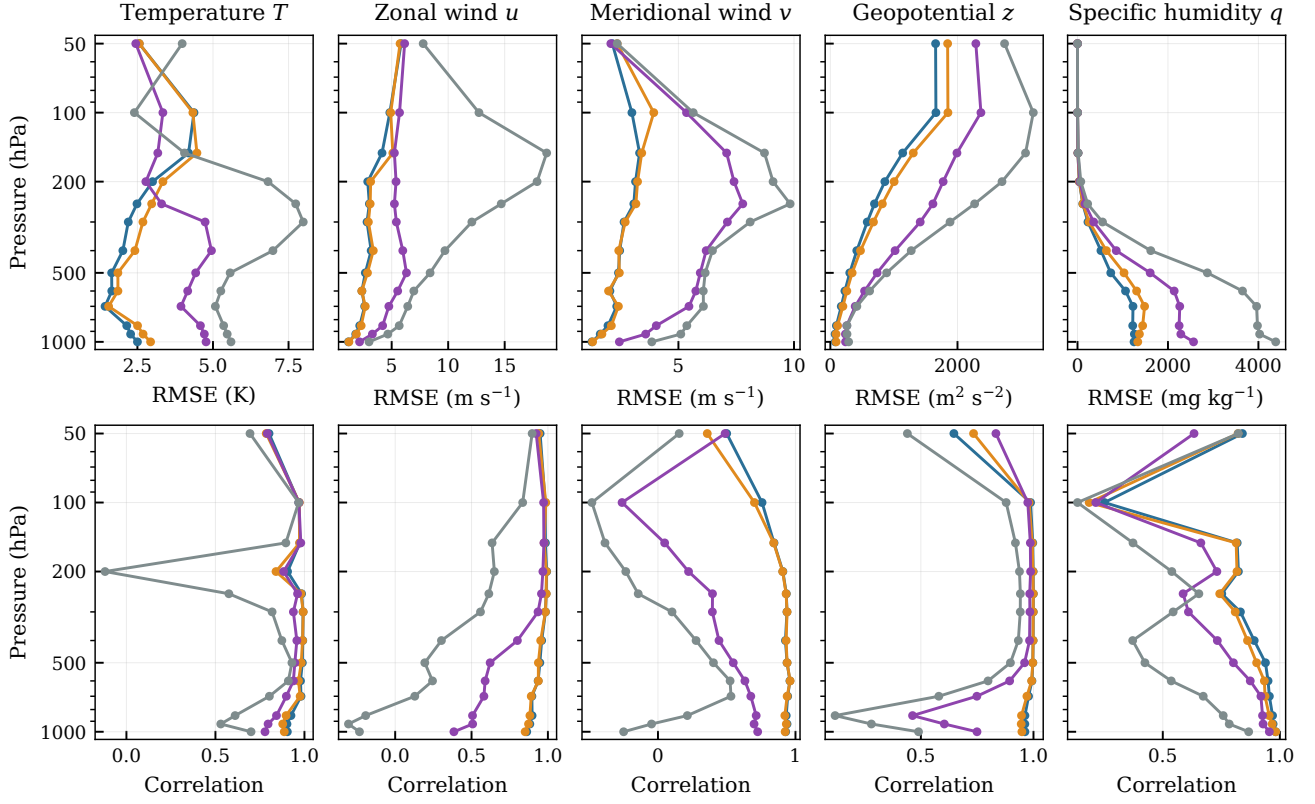


Figure 10: As Fig. 5, for the RO ablation experiments.

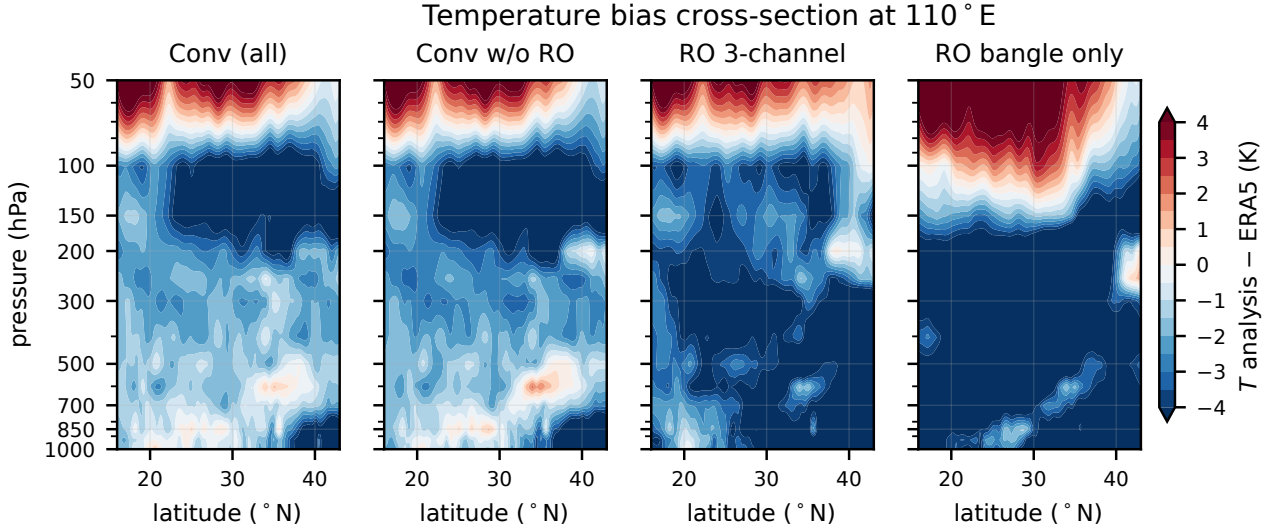


Figure 11: Temperature analysis-minus-ERA5 cross-sections at 110°E for the four RO-ablation experiments. The bending-angle-only analysis (right) develops a large cold bias below 100 hPa, while the RO three-channel analysis remains structurally close to the full conventional stream in the upper troposphere and lower stratosphere.

and the model is only valid for background statistics from its training system.

8 Practical implications

For accelerator porting. The MPS defect documented here produces *plausible but degraded*

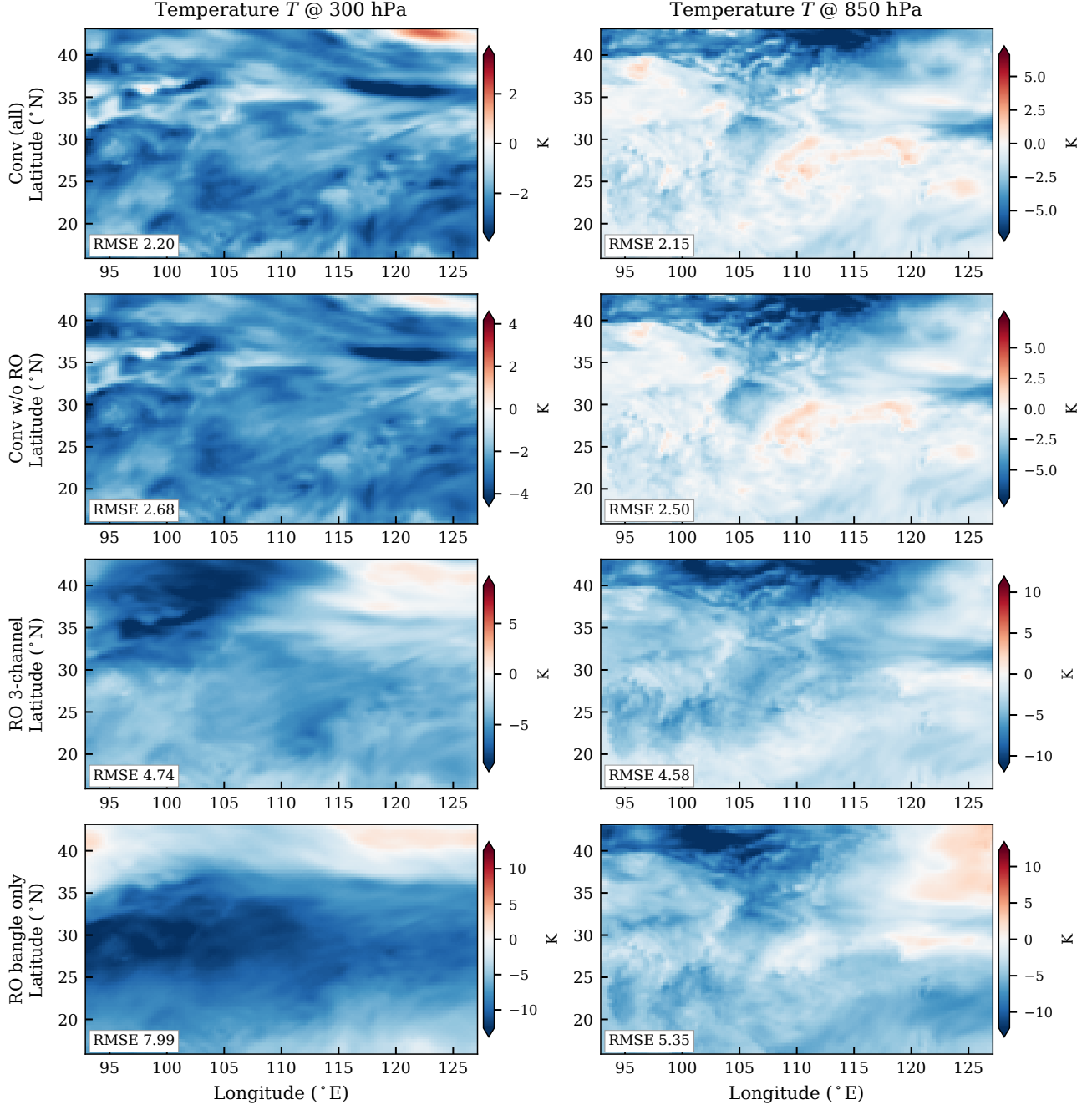


Figure 12: Analysis-minus-ERA5 temperature differences at 300 hPa (left) and 850 hPa (right) for the four ablation experiments.

analyses and is invisible to smoke tests: at 2×10^4 observations the encoder output is bit-consistent across devices. The general lesson for porting ML DA systems is that parity testing must be performed at production observation scale, on every input tensor of the data pipeline (not only the model), and against field-level truth, because an ML DA system degrades gracefully and therefore

hides memory-safety bugs. The two patches presented here are each about twenty lines and make no change to the model or its weights; they are generic enough to be reusable for other physics-memo DiT models on Metal.

For re-using HealDA with new observation sources. The checkpoint’s observation embed-

ders are frozen: the bucket membership (five families) and bin counts are hard-coded in the trained weights, so *adding* a sensor family requires training a new embedder; *subsetting* existing families, however, is natively supported through the observability masks, as our ablation demonstrates. For a group wishing to feed, e.g., FengYun-3 GNOS bending angles (Liao et al. 2016) into the released checkpoint, the practical path is the `conv` bucket’s `gps` slots, with two hard requirements: (i) the companion background T/q values must be provided from a background consistent with the training system’s climatology, since the ablation shows they carry decisive information; and (ii) the bending-angle normalization (training mean 0.004, 11 rad, standard deviation 0.006, 02 rad) must be applied unchanged. Vertical coverage is additionally bounded by the training data (RO reports to 0.5 hPa in the UFS stream); systematically truncated profiles constitute an out-of-distribution sparsity pattern that the observability masks will flag but cannot correct.

For laptop-scale DA experiments. The repaired MPS pipeline completes the conventional-only analysis in 59s (8.6 GB peak) and the conventional-plus-satellite analysis in 406s (10.5 GB peak) on a fanless-class laptop (Table 3), against 960s and 16.0 GB on CPU. This makes controlled ablation ensembles of the kind in §6 an interactive rather than batch activity, and puts a single-window global analysis within reach of classroom-scale hardware.

9 Conclusions

We ported the HealDA ML data assimilation system to Apple Silicon, diagnosed and repaired a silent memory-corruption defect that had previously been misattributed to reduced-precision arithmetic, and used the repaired pipeline to quantify, in a controlled single-case setting, the analysis impact of microwave radiances and of GNSS radio occultation observations over East Asia. The principal findings are that (i) satellite radiances dominate the analysis quality, reducing layer-mean RMSE by 20–78% depending

on the variable and largely removing a zonal-mean mass-field bias; (ii) RO observations, although 71% of the conventional stream by count, contribute a further 10–15% RMSE reduction concentrated in temperature, geopotential and humidity, with RO-only skill matching the full conventional stream in the upper stratosphere; and (iii) the RO analysis skill depends jointly on the bending angles and their background-value companion channels, consistent with a learned $(O, B) \rightarrow x_a$ map rather than a retrieval, and illustrating how a trained network internalizes the roles of B , R and the adjoint of a variational system.

Limitations. Our verification is a single case (2024-06-01 00 UTC) over East Asia against ERA5, which is itself the training target of HealDA; neither sampling uncertainty nor the verification-target relationship is addressed, and the RMSE gaps reported here should be read as process-level attribution rather than climatological skill estimates. The MPS repair is validated to bf16 noise, not to fp32, and the Metal defect itself is upstream of our code. Extending the ablation to multiple dates, to the satellite bucket, and to observation-system experiments with real FengYun-3 GNOS profiles is the natural continuation.

A Reproducing the experiments

The repository layout is `code/` (pipeline scripts), `paper/` (this manuscript, figure script and statistics) and `analysis_outputs/` (regional gridded analyses in CF-1.8 NetCDF, openable with `ncview`). Environment: Python 3.12, PyTorch 2.10, `earth2studio` 0.18.0, `physicsnemo` 2.2.2 (`uv` lock file included). The HealDA checkpoint (`healda_ufs_era5.mdlus`) is fetched automatically by `earth2studio` on first use. Each experiment in this paper is a single invocation of the analysis driver with a date, a device (`--device cpu` or `--device mps`), an output path, and a sensor selection: `--sensors conv` for the conventional stream, `--sensors both` to add microwave radiances, and `--conv-vars` to restrict the conventional bucket to an arbitrary subset of its eight channels (e.g. the three RO channels alone, or the conventional stream with RO removed)—the

switch that enables the ablations of §6.

The `--device mps` path applies both patches of §3 automatically. Native HEALPix outputs are converted to regular lat-lon CF-1.8 files by `code/06_to_latlon.py` (cKDTree $k = 6$ Gaussian-weighted regridding, 0.55° effective smoothing) and verified by `paper/make_figures.py`, which recomputes every statistic and figure in this manuscript from the NetCDF files. Note that the verification figures use direct Delaunay interpolation of the native grid (no smoothing), which is the statistically conservative choice; the smoothed regridding is provided for visualization.

B Per-level verification statistics

The complete per-level table (7 experiments $\times$ 5 variables $\times$ 13 levels, with RMSE, bias, correlation and standard-deviation ratio against ERA5) is machine-readable at `paper/stats.csv` and `paper/stats.json` in the repository. Figure 10 and Figure 5 display the same data graphically.

Data and code availability

All scripts (analysis driver, MPS patches, HEALPix shim, regridding and figure generation), the per-level verification statistics (`stats.csv`), and this manuscript are available at <https://github.com/sonet-hkust/healda-apple-silicon>. The HealDA checkpoint is distributed with `earth2studio`; ERA5 data were obtained from the Copernicus Climate Data Store.

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